

EcoPaXX® Q-KS

PA410 FR

Flame Retardant (halogen- phosphorous-free)

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EcoPaXX® Q-KS is a versatile halogen-free flame-retarded long aliphatic polyamide with UL94 V0 & CTI 600V rating with high flow and ductility. It offers excellent retention of electric properties upon humidity uptake.

Sustainability

Bio-based – 14C measurable

Properties	Typical Data	Unit	Test Method
Rheological properties	dry / cond		
Molding shrinkage (parallel)	1.1 / *	%	ISO 294-4
Molding shrinkage (normal)	0.9 / *	%	ISO 294-4
Mechanical properties	dry / cond		
Tensile modulus	3800 / 2200	MPa	ISO 527-1/-2
Nominal strain at break	6 / 30	%	ISO 527-1/-2
Yield stress	75 / 50	MPa	ISO 527-1/-2
Yield strain	4 / 12.5	%	ISO 527-1/-2
Tensile modulus (80°C)	1300 / -	MPa	ISO 527-1/-2
Charpy impact strength (+23°C)	75 / -	kJ/m ²	ISO 179/1eU
Charpy impact strength (-30°C)	85 / 85	kJ/m ²	ISO 179/1eU
Charpy notched impact strength (+23°C)	3 / 5	kJ/m ²	ISO 179/1eA
Charpy notched impact strength (-30°C)	2.8 / 2.7	kJ/m ²	ISO 179/1eA
Flexural modulus	3900 / 2300	MPa	ISO 178
Flexural strength	120 / 65	MPa	ISO 178
Thermal properties	dry / cond		
Melting temperature (10°C/min)	250 / *	°C	ISO 11357-1/-3

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Properties	Typical Data	Unit	Test Method
Temp. of deflection under load (1.80 MPa)	110 / *	°C	ISO 75-1/-2
Temp. of deflection under load (0.45 MPa)	190 / *	°C	ISO 75-1/-2
Burning Behav. at 1.5 mm nom. thickn.	V-0 / *	class	IEC 60695-11-10
Thickness tested	1.5 / *	mm	IEC 60695-11-10
Burning Behav. at 3.0 mm nom. thickn.	V-0 / *	class	IEC 60695-11-10
Thickness tested	3 / *	mm	IEC 60695-11-10
Oxygen index	36 / *	%	ISO 4589-1/-2
Glow Wire Flammability Index GWFI	960 / -	°C	IEC 60695-2-12
GWFI (Thickness (1) tested)	0.75 / -	mm	IEC 60695-2-12
Glow Wire Ignition Temperature GWIT	960 / -	°C	IEC 60695-2-13
GWIT (Thickness (1) tested)	0.75 / -	mm	IEC 60695-2-13

Electrical properties

dry / cond

Relative permittivity (100Hz)	3.85 / -	-	IEC 62631-2-1
Relative permittivity (1 MHz)	3.45 / -	-	IEC 62631-2-1
Dissipation factor (100 Hz)	140 / -	E-4	IEC 62631-2-1
Dissipation factor (1 MHz)	200 / -	E-4	IEC 62631-2-1
Volume resistivity	>1E13 / 7E11	Ohm*m	IEC 62631-3-1
Surface resistivity	- / 6E13	Ohm	IEC 62631-3-2
Electric strength	32 / 27	kV/mm	IEC 60243-1
Comparative tracking index	600 / 600	V	IEC 60112

Other properties

dry / cond

Water absorption	4.8 / *	%	Sim. to ISO 62
Water absorption in water at 23°C after 24h	0.6 / *	%	ISO 62
Humidity absorption	1.8 / *	%	Sim. to ISO 62
Density	1160 / -	kg/m ³	ISO 1183
Biobased content (in polymer)	68	% (Bio C/Total C)	ASTM D6866-12 Method B

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